COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/22/2026	**DATE SUBMITTED:**	10/22/2026
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ACTIVITY NO. 12.1

Build a Sample Web App in a Docker Container

1. PROCEDURE OUTPUT

Part 2

Step 1:

```
[analyst@secOps ~]$ mkdir -p ~/labs/devnet-src/sample-app
[analyst@secOps ~]$ cd ~/labs/devnet-src/sample-app
[analyst@secOps sample-app]$ touch user-input.sh
[analyst@secOps sample-app]$
```

Step 2:

```
[analyst@secOps sample-app]$ nano user-input.sh
[analyst@secOps sample-app]$ bash user-input.sh
Enter Your Name:
Angelo
YOur name is Angelo.
[analyst@secOps sample-app]$
```

Step 3-4:

```
[analyst@secOps sample-app]$ cat user-input.sh
#!bin/bash

echo "Enter Your Name:"
read userName
echo "YOur name is $userName."
```

Step 7:

```
[analyst@secOps sample-app]$ ls -l user-input.sh
-rw-r--r-- 1 analyst analyst 83 Oct 21 20:59 user-input.sh
[analyst@secOps sample-app]$ chmod a+x user-input.sh
[analyst@secOps sample-app]$ ls -l user-input.sh
-rwxr-xr-x 1 analyst analyst 83 Oct 21 20:59 user-input.sh
[analyst@secOps sample-app]$
```

Step 8:

```
[analyst@secOps sample-app]$ mv user-input.sh user-input
[analyst@secOps sample-app]$
```

Part 3

Step 1:

```
[analyst@secOps sample-app]$ pip3 install flask
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: flask in /usr/lib/python3.6/site-packages (2.0.3)
Requirement already satisfied: Werkzeug>=2.0 in /usr/lib/python3.6/site-packages (from flask) (2.0.3)
Requirement already satisfied: itsdangerous>=2.0 in /usr/lib/python3.6/site-packages (from flask) (2.0.1)
Requirement already satisfied: click>=7.1.2 in /usr/lib/python3.6/site-packages (from flask) (8.0.4)
Requirement already satisfied: Jinja2>=3.0 in /usr/lib/python3.6/site-packages (from flask) (3.0.3)
Requirement already satisfied: importlib-metadata in /usr/lib/python3.6/site-packages (from click>=7.1.2->fla
Requirement already satisfied: MarkupSafe>=2.0 in /usr/lib/python3.6/site-packages (from Jinja2>=3.0->flask)
Requirement already satisfied: dataclasses in /usr/lib/python3.6/site-packages (from Werkzeug>=2.0->flask) (0
Requirement already satisfied: zipp>=0.5 in /usr/lib/python3.6/site-packages (from importlib-metadata->click>
Requirement already satisfied: typing-extensions>=3.6.4 in /usr/lib/python3.6/site-packages (from importlib-m
flask) (4.1.1)
```

Step 2-4:

```
[analyst@secOps sample-app]$ nano sample_app.py
[analyst@secOps sample-app]$ cat sample_app.py
from flask import Flask
from flask import request

sample = Flask(__name__)

[analyst@secOps sample-app]$
```

Step 5:

```
[analyst@secOps sample-app]$ cat sample_app.py
from flask import Flask
from flask import request

sample = Flask(__name__)

@sample.route("/")
def main():
    return "You are calling me from " + request.remote_addr + "\n"


[analyst@secOps sample-app]$
```

Step 6:

```
[analyst@secOps sample-app]$ nano sample_app.py
[analyst@secOps sample-app]$ cat sample_app.py
from flask import Flask
from flask import request

sample = Flask(__name__)

@sample.route("/")
def main():
    return "You are calling me from " + request.remote_addr + "\n"

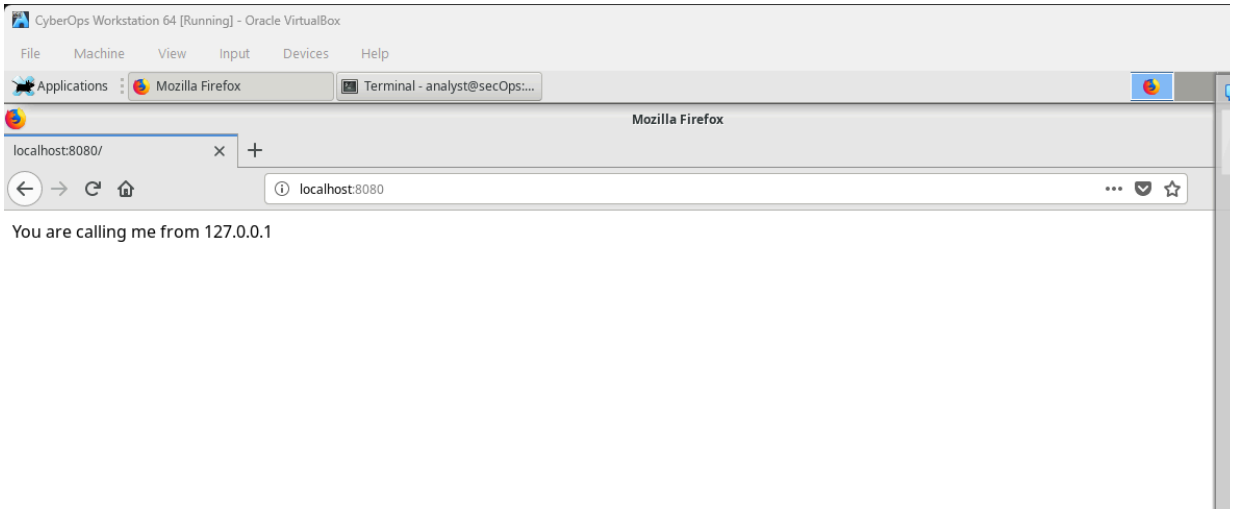
if __name__ == "__main__":
    sample.run(host="0.0.0.0", port=8080)

[analyst@secOps sample-app]$
```

Step 7:

```
[analyst@secOps sample-app]$ python3 sample_app.py
 * Serving Flask app 'sample_app' (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deployment.
   Use a production WSGI server instead.
 * Debug mode: off
 * Running on all addresses.
   WARNING: This is a development server. Do not use it in a production deployment.
 * Running on http://10.6.12.160:8080/ (Press CTRL+C to quit)
```

Step 8:



```
^C[analyst@secOps sample-app]$ python3 sample_app.py
 * Serving Flask app 'sample_app' (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deployment.
   Use a production WSGI server instead.
 * Debug mode: off
 * Running on all addresses.
   WARNING: This is a development server. Do not use it in a production deployment.
 * Running on http://10.6.12.160:8080/ (Press CTRL+C to quit)
127.0.0.1 - - [21/Oct/2026 21:18:35] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [21/Oct/2026 21:18:35] "GET /favicon.ico HTTP/1.1" 404 -
^C[analyst@secOps sample-app]$
```

Part 4

Step 1:

```
[analyst@secOps sample-app]$ nano templates/index.html
[analyst@secOps sample-app]$ cat templates/index.html
<html>
<head>
    <title>Sample app</title>
    <link rel="stylesheet" href="/static/style.css" />
</head>
<body>
    <h1>YOu are calling me from {{request.remote_addr}}</h1>
</body>
</html>

[analyst@secOps sample-app]$
```

```
[analyst@secOps sample-app]$ cat static/style.css
body {background: lightsteelblue;}

[analyst@secOps sample-app]$
```

Step 2-5:

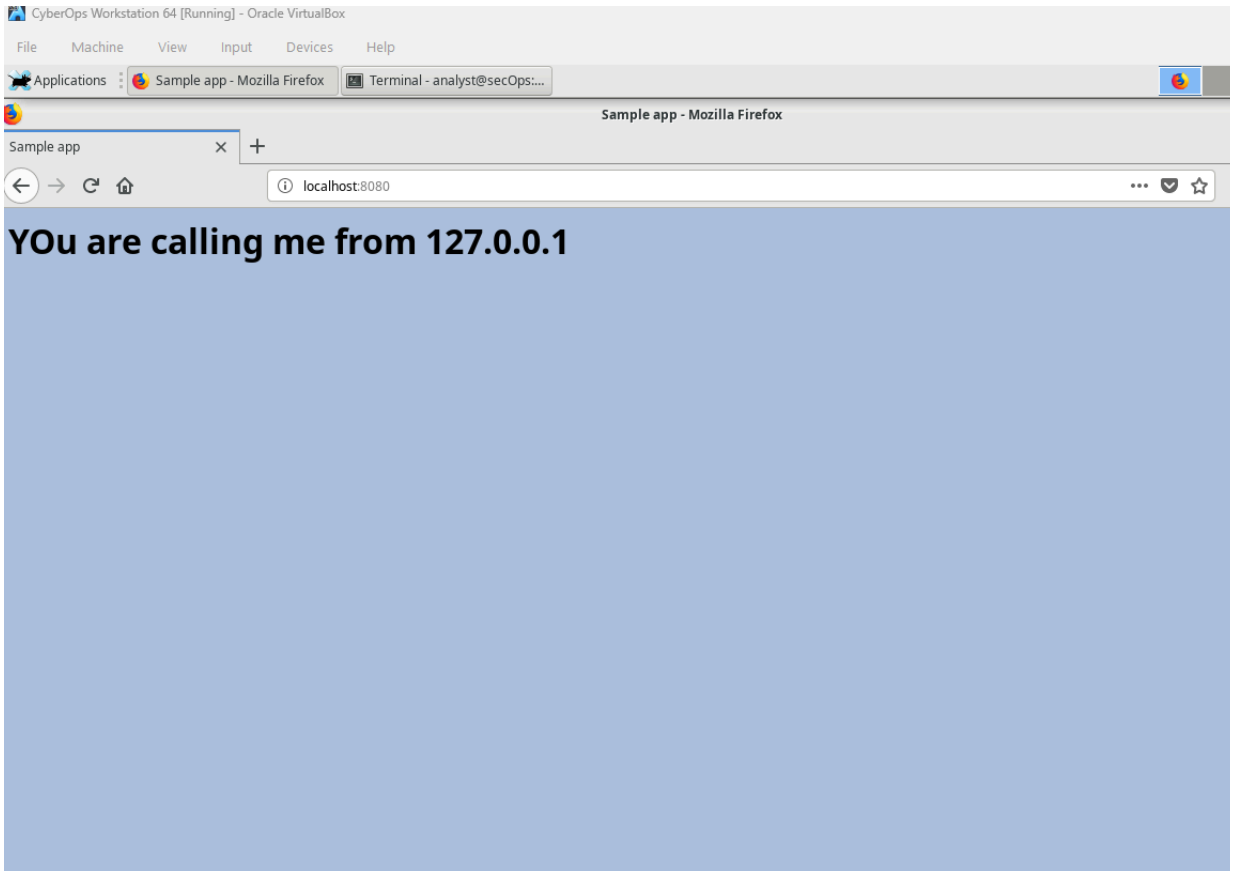
```
[analyst@secOps sample-app]$ nano sample_app.py
[analyst@secOps sample-app]$ cat sample_app.py
from flask import Flask
from flask import request
from flask import render_template

sample = Flask(__name__)

@sample.route("/")
def main():
    return "You are calling me from " + request.remote_addr + "\n"

if __name__ == "__main__":
    sample.run(host="0.0.0.0", port=8080)

[analyst@secOps sample-app]$
```



Part 5

Step 1:

```
[analyst@secOps sample-app]$ nano sample-app.sh
[analyst@secOps sample-app]$ cat sample-app.sh
#!/bin/bash

mkdir tempdir
mkdir tempdir/templates
mkdir tempdir/static

[analyst@secOps sample-app]$
```

```
[analyst@secOps sample-app]$ chmod +x sample-app.sh
[analyst@secOps sample-app]$
```

Step 2:

```
[analyst@secOps sample-app]$ nano sample-app.sh
[analyst@secOps sample-app]$ cat sample-app.sh
#!/bin/bash

mkdir tempdir
mkdir tempdir/templates
mkdir tempdir/static

cp sample_app.py tempdir/.
cp -r templates/* tempdir/templates/.
cp -r static/* tempdir/static/.
```

Step 3:

```
[analyst@secOps sample-app]$ cat sample-app.sh
#!/bin/bash

mkdir tempdir
mkdir tempdir/templates
mkdir tempdir/static

cp sample_app.py tempdir/.
cp -r templates/* tempdir/templates/.
cp -r static/* tempdir/static/.

echo "FROM python" >> tempdir/Dockerfile
echo "RUN pip install flask" >> tempdir/Dockerfile
echo "COPY ./static /home/myapp/static/" >> tempdir/Dockerfile
echo "COPY ./templates /home/myapp/templates/" >> tempdir/Dockerfile
echo "COPY sample_app.py /home/myapp/" >> tempdir/Dockerfile
echo "EXPOSE 8080" >> tempdir/Dockerfile
echo "CMD python3 /home/myapp/sample_app.py" >> tempdir/Dockerfile


[analyst@secOps sample-app]$
```

Step 5:

```
analyst@secOps sample-app]$ nano sample-app.sh
analyst@secOps sample-app]$ cat sample-app.sh
!/bin/bash

kdir tempdir
kdir tempdir/templates
kdir tempdir/static

p sample_app.py tempdir/.
p -r templates/* tempdir/templates/.
p -r static/* tempdir/static/.

cho "FROM python" >> tempdir/Dockerfile
cho "RUN pip install flask" >> tempdir/Dockerfile
cho "COPY ./static /home/myapp/static/" >> tempdir/Dockerfile
cho "COPY ./templates /home/myapp/templates/" >> tempdir/Dockerfile
cho "COPY sample_app.py /home/myapp/" >> tempdir/Dockerfile
cho "EXPOSE 8080" >> tempdir/Dockerfile
cho "CMD python3 /home/myapp/sample_app.py" >> tempdir/Dockerfile

d tempdir
ocker build -t sampleapp
```

Step 6:

```
[analyst@secOps sample-app]$ nano sample-app.sh
[analyst@secOps sample-app]$ cat sample-app.sh
#!/bin/bash

mkdir tempdir
mkdir tempdir/templates
mkdir tempdir/static

cp sample_app.py tempdir/.
cp -r templates/* tempdir/templates/.
cp -r static/* tempdir/static/.

echo "FROM python" >> tempdir/Dockerfile
echo "RUN pip install flask" >> tempdir/Dockerfile
echo "COPY ./static /home/myapp/static/" >> tempdir/Dockerfile
echo "COPY ./templates /home/myapp/templates/" >> tempdir/Dockerfile
echo "COPY sample_app.py /home/myapp/" >> tempdir/Dockerfile
echo "EXPOSE 8080" >> tempdir/Dockerfile
echo "CMD python3 /home/myapp/sample_app.py" >> tempdir/Dockerfile

cd tempdir
docker build -t sampleapp

docker run -t -d -p 8080:8080 --name samplerunning sampleapp
docker ps -a

[analyst@secOps sample-app]$
```

Part 6

Step 1:

```
[analyst@secOps sample-app]$ bash ./sample-app.sh
mkdir: cannot create directory ‘tempdir’: File exists
mkdir: cannot create directory ‘tempdir/templates’: File exists
mkdir: cannot create directory ‘tempdir/static’: File exists
./sample-app.sh: line 20: docker: command not found
./sample-app.sh: line 22: docker: command not found
./sample-app.sh: line 23: docker: command not found
[analyst@secOps sample-app]$
```

Step 2:

```
[analyst@secOps sample-app]$ ls tempdir/
Dockerfile  sample_app.py  static  templates
[analyst@secOps sample-app]$ cat tempdir/Dockerfile
FROM python
RUN pip install flask
COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/
EXPOSE 8080
CMD python3 /home/myapp/sample_app.py
FROM python
RUN pip install flask
COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/
EXPOSE 8080
CMD python3 /home/myapp/sample_app.py
[analyst@secOps sample-app]$
```

Step 3:

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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